

PSG COLLEGE OF TECHNOLOGY
DEPARTMENT OF COMPUTER APPLICATIONS
I SEMESTER MCA
PROBLEM SHEET ON ARRAYS

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1. Write a program to accept the integer values and display the second largest value in an array.

```
import java.util.Arrays;

import java.util.Scanner;

public class SecondLargestInArray {

    public static void main(String[] args) {

        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter the number of elements: ");

        int n = scanner.nextInt();

        int[] arr = new int[n];

        System.out.println("Enter the elements:");

        for (int i = 0; i < n; i++) {

            arr[i] = scanner.nextInt();

        }

        Arrays.sort(arr);

        // Assuming at least 2 elements, find second largest (handle
        duplicates by finding unique)

        int largest = arr[n - 1];

        int secondLargest = -1;
```

```

        for (int i = n - 2; i >= 0; i--) {

            if (arr[i] != largest) {

                secondLargest = arr[i];

                break;

            }

        }

        if (secondLargest == -1) {

            System.out.println("No second largest element.");

        } else {

            System.out.println("Second largest: " + secondLargest);

        }

    }

}

```

2. Write a program to sort the list of numbers in an ascending and descending order.

```

import java.util.Arrays;

import java.util.Collections;

import java.util.Scanner;

public class SortArray {

    public static void main(String[] args) {

        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter the number of elements: ");
    }
}

```

```

        int n = scanner.nextInt();

        Integer[] arr = new Integer[n];

        System.out.println("Enter the elements:");

        for (int i = 0; i < n; i++) {

            arr[i] = scanner.nextInt();

        }

        // Ascending

        Arrays.sort(arr);

        System.out.println("Ascending: " + Arrays.toString(arr));

        // Descending

        Arrays.sort(arr, Collections.reverseOrder());

        System.out.println("Descending: " + Arrays.toString(arr));

    }
}

```

3. Write a program to search for a specified number in an array and display with its position.

```

import java.util.Arrays;

import java.util.Collections;

import java.util.Scanner;

public class SortArray {

    public static void main(String[] args) {

        Scanner scanner = new Scanner(System.in);
    }
}

```

```

        System.out.print("Enter the number of elements: ");

        int n = scanner.nextInt();

        Integer[] arr = new Integer[n];

        System.out.println("Enter the elements:");

        for (int i = 0; i < n; i++) {

            arr[i] = scanner.nextInt();

        }

        // Ascending

        Arrays.sort(arr);

        System.out.println("Ascending: " + Arrays.toString(arr));

        // Descending

        Arrays.sort(arr, Collections.reverseOrder());

        System.out.println("Descending: " + Arrays.toString(arr));

    }
}

```

4. Write a program to find the occurrence of positive, negative, even and odd elements for a given array.

```

import java.util.Scanner;

public class CountArrayElements {

    public static void main(String[] args) {

        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter the number of elements: ");
    }
}

```

```

int n = scanner.nextInt();

int[] arr = new int[n];

System.out.println("Enter the elements:");

for (int i = 0; i < n; i++) {

    arr[i] = scanner.nextInt();

}

int positive = 0, negative = 0, even = 0, odd = 0;

for (int num : arr) {

    if (num > 0) positive++;

    if (num < 0) negative++;

    if (num % 2 == 0) even++;

    if (num % 2 != 0) odd++;

}

System.out.println("Positive: " + positive);

System.out.println("Negative: " + negative);

System.out.println("Even: " + even);

System.out.println("Odd: " + odd);

}
}

```

5. In a gymnastics or diving competition, a contestant's score is calculated by dropping the lowest and highest scores and then adding the remaining scores. Write a program that allows the user to enter eight judges' scores and then outputs the point received by the contestant. A judge awards point between 1 and 10, with 1 being the lowest and 10 being the highest. For example, if the scores are: 9.2, 9.3, 9.0, 9.9, 9.5, 9.5, 9.6 and 9.8, then the

contestant receives a total of 56.9 points.

Here is the sample output.

Enter 8 scores out of ten points :

9.1 9.0 8.9 8.8 9.4 7.9 8.6 9.8

Your lowest score is 7.90

Your maximum score is 9.80

Your total point is 53.80

Your average point is 8.97

```
import java.util.Arrays;

import java.util.Scanner;

public class GymnasticsScore {

    public static void main(String[] args) {

        Scanner scanner = new Scanner(System.in);

        double[] scores = new double[8];

        System.out.println("Enter 8 scores out of ten points:");

        for (int i = 0; i < 8; i++) {

            scores[i] = scanner.nextDouble();

        }

        Arrays.sort(scores);

        double min = scores[0];

        double max = scores[7];

        double total = 0;

        for (int i = 0; i < 8; i++) {

            total += scores[i];

        }

        double average = total / 8;

        System.out.printf("Your lowest score is %.2f\n", min);
```

```

        System.out.printf("Your maximum score is %.2f%n", max);

        System.out.printf("Your total point is %.2f%n", total);

        System.out.printf("Your average point is %.2f%n", average);

    }
}

```

6. Write a program that accepts an array and a key value. Rotate the array element by 'key' times.

Example:

Input: arr[] =

key=2 Output : [3

5, 6, 1, 2]

```

import java.util.Arrays;
import java.util.Scanner;

public class RotateArray {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter the number of elements: ");
        int n = scanner.nextInt();
        int[] arr = new int[n];
        System.out.println("Enter the elements:");
        for (int i = 0; i < n; i++) {
            arr[i] = scanner.nextInt();
        }
        System.out.print("Enter the key (rotation times): ");
        int key = scanner.nextInt() % n; // Handle large key
        int[] rotated = new int[n];
        for (int i = 0; i < n; i++) {
            rotated[i] = arr[(i + key) % n];
        }
        System.out.println("Rotated array: " +
Arrays.toString(rotated));
    }
}

```

}